
Math 335 Homework 1 Solutions

Release 0.1

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0.1 Submitted Homework Assignment MATH335 HW1

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0.2 Solution of Problem 1:

Equation	Order	Type	Reason
(a)	1	Nonlinear	Contains $\sin(y)$
(b)	1	Linear	y and y' appear linearly
(c)	3	Linear	All terms are linear in y and its derivatives
(d)	3	Nonlinear	Contains y^4
(e)	1	Nonlinear	Contains $1/y$
(f)	1	Nonlinear	Contains y^2

0.3 Solution of Problem 2(a) and (b)

0.3.1 Solution 2(a)

Given

$$y' = y^2 - 1, \quad y(0) = 0$$

- Equilibrium points:

$$y^2 - 1 = 0 \Rightarrow y = \pm 1$$

- Sign of y' :

- $y > 1 \Rightarrow y' > 0$ (solution moves upward)
- $-1 < y < 1 \Rightarrow y' < 0$ (solution moves downward)
- $y < -1 \Rightarrow y' > 0$ (solution moves upward)

- Stability:

- $y = -1$ is a stable equilibrium.
- $y = 1$ is an unstable equilibrium.

- Since $y(0) = 0$ lies in $(-1, 1)$, we have $y' < 0$, so the solution decreases toward the stable equilibrium $y = -1$.

Therefore,

$$\lim_{t \rightarrow \infty} y(t) = -1.$$

For $y' = y^2 - 1$:

Equilibrium points:

$$y^2 - 1 = 0$$

$$y = \pm 1$$

Representative slopes:

y	y'
2	3
1	0
0	-1
-1	0
-2	3

Thus:

- $y > 1 \Rightarrow y' > 0$ (solutions move upward)
- $-1 < y < 1 \Rightarrow y' < 0$ (solutions move downward)
- $y < -1 \Rightarrow y' > 0$ (solutions move upward)

Since $y(0)=0$ lies in $(-1,1)$, the solution decreases and approaches $y=-1$ as $t \rightarrow \infty$.

Slope field plot

0.3.2 Solution 2(b)

Given

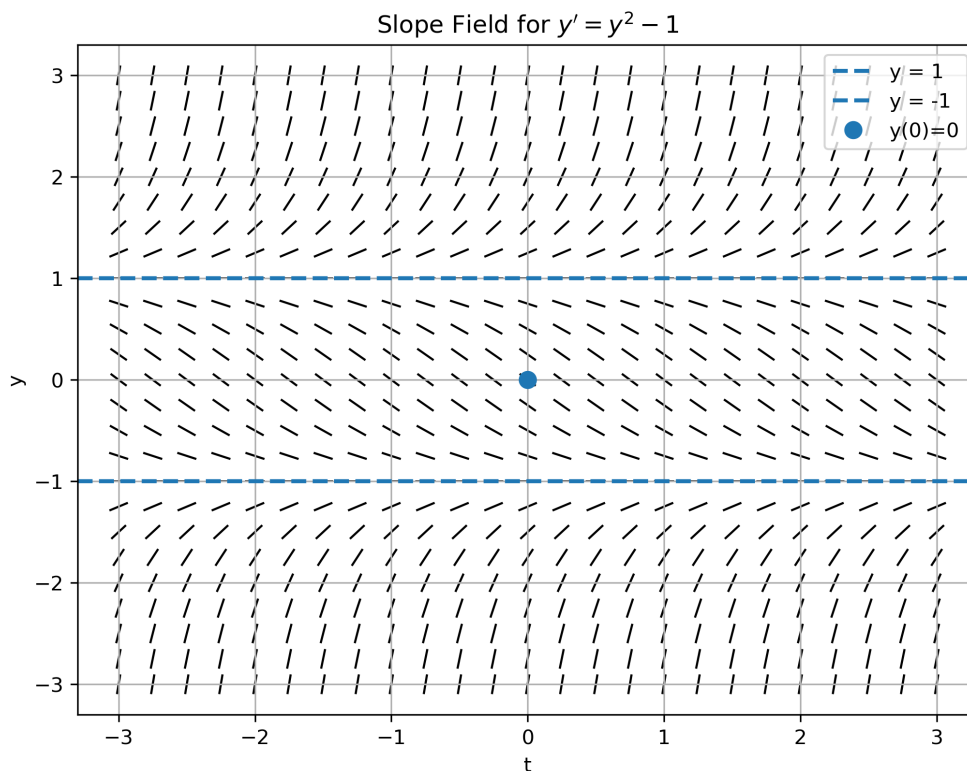
$$y' = t^2 - 1, \quad y(0) = 0$$

Representative slopes:

t	y'
-2	3
-1	0
0	-1
1	0
2	3

Sign of y' :

- $t < -1 \Rightarrow y' > 0$ (upward slopes)
- $t = -1 \Rightarrow y' = 0$ (horizontal slopes)
- $-1 < t < 1 \Rightarrow y' < 0$ (downward slopes)
- $t = 1 \Rightarrow y' = 0$ (horizontal slopes)
- $t > 1 \Rightarrow y' > 0$ (upward slopes)

Fig. 1: Slope field for $y' = y^2 - 1$.

Since $y(0) = 0$, the solution initially decreases because

$$y'(0) = 0^2 - 1 = -1.$$

For large t ,

$$y' = t^2 - 1 > 0,$$

so the solution eventually turns upward and increases without bound.

Therefore,

$$\lim_{t \rightarrow \infty} y(t) = \infty.$$

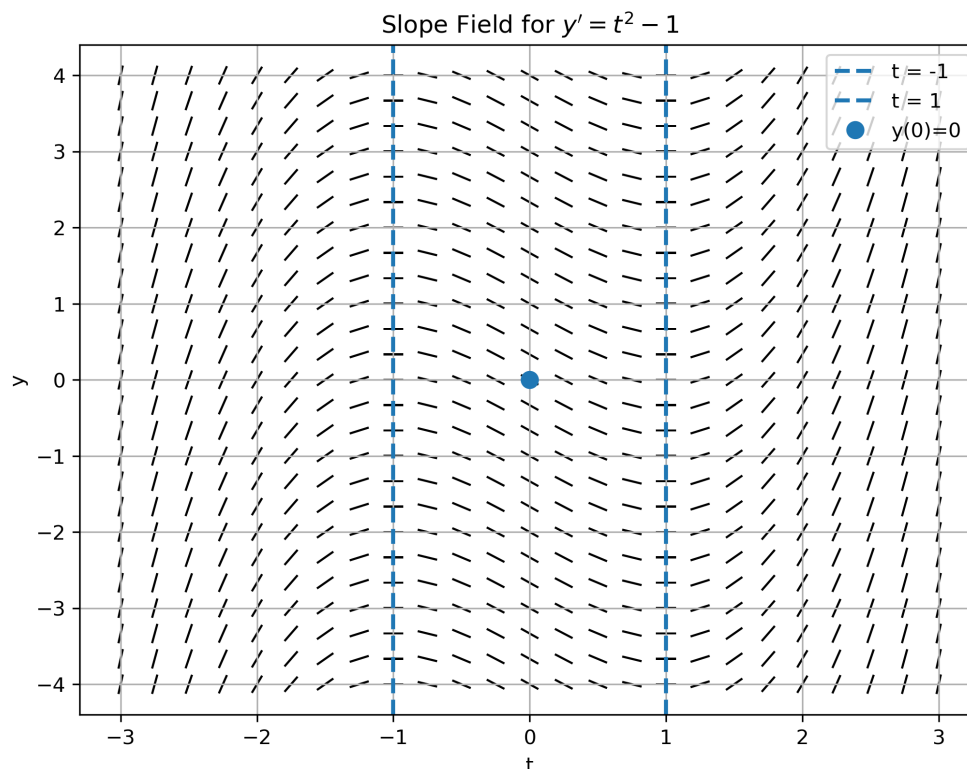
Slope Field plot

0.4 Solution of Problem 3

Question: Solve the initial value problem

$$y' = \frac{1}{t^2 - 3t + 2}, \quad y(0) = 1.$$

Answer:

Fig. 2: Slope field for $y' = t^2 - 1$.

0.4.1 Step 1: Rewrite the Differential Equation

Recall that

$$y' = \frac{dy}{dt}.$$

Therefore,

$$\frac{dy}{dt} = \frac{1}{t^2 - 3t + 2}.$$

To recover the function $y(t)$, we integrate both sides with respect to t .

$$y = \int \frac{1}{t^2 - 3t + 2} dt + C.$$

The main task is therefore to evaluate the integral

$$\int \frac{1}{t^2 - 3t + 2} dt.$$

0.4.2 Step 2: Factor the Denominator

Before using partial fractions, factor the quadratic denominator.

We seek two numbers whose

- product is 2,
- sum is -3 .

These numbers are -1 and -2 .

Therefore,

$$t^2 - 3t + 2 = (t - 1)(t - 2).$$

Substituting this factorization into the integral gives

$$y = \int \frac{1}{(t - 1)(t - 2)} dt + C.$$

0.4.3 Step 3: Set Up the Partial Fraction Decomposition

Because the denominator consists of two distinct linear factors, we assume that

$$\frac{1}{(t - 1)(t - 2)} = \frac{A}{t - 1} + \frac{B}{t - 2},$$

where A and B are constants to be determined.

0.4.4 Step 4: Eliminate the Denominators

Multiply both sides by the common denominator

$$(t - 1)(t - 2).$$

This gives

$$(t - 1)(t - 2) \left(\frac{1}{(t - 1)(t - 2)} \right) = (t - 1)(t - 2) \left(\frac{A}{t - 1} + \frac{B}{t - 2} \right).$$

Simplifying each side,

Left-hand side:

$$1.$$

Right-hand side:

$$A(t - 2) + B(t - 1).$$

Hence,

$$1 = A(t - 2) + B(t - 1).$$

This equation must be true for every value of t .

0.4.5 Step 5: Find the Constant A

Choose

$$t = 1.$$

Substituting into

$$1 = A(t - 2) + B(t - 1)$$

gives

$$1 = A(1 - 2) + B(1 - 1).$$

Simplify:

$$1 = A(-1) + B(0).$$

$$1 = -A.$$

Therefore,

$$A = -1.$$

0.4.6 Step 6: Find the Constant B

Choose

$$t = 2.$$

Substituting into

$$1 = A(t - 2) + B(t - 1)$$

gives

$$1 = A(2 - 2) + B(2 - 1).$$

Simplify:

$$1 = A(0) + B(1).$$

$$1 = B.$$

Therefore,

$$B = 1.$$

0.4.7 Step 7: Write the Partial Fraction Expansion

Substituting

$$A = -1, \quad B = 1,$$

into the decomposition gives

$$\frac{1}{(t-1)(t-2)} = -\frac{1}{t-1} + \frac{1}{t-2}.$$

Therefore,

$$y = \int \left(-\frac{1}{t-1} + \frac{1}{t-2} \right) dt + C.$$

0.4.8 Step 8: Integrate the First Term

Consider

$$\int -\frac{1}{t-1} dt.$$

Let

$$u = t - 1.$$

Differentiate both sides:

$$\frac{du}{dt} = \frac{d}{dt}(t - 1).$$

Since

$$\frac{d}{dt}(t) = 1, \quad \frac{d}{dt}(-1) = 0,$$

we obtain

$$\frac{du}{dt} = 1.$$

Multiplying by dt gives

$$du = dt.$$

Therefore,

$$\int -\frac{1}{t-1} dt = -\int \frac{1}{u} du.$$

Using the logarithmic integration rule

$$\int \frac{1}{u} du = \ln |u|,$$

we obtain

$$-\ln |u|.$$

Substituting back $u = t - 1$,

$$-\ln |t - 1|.$$

0.4.9 Step 9: Integrate the Second Term

Consider

$$\int \frac{1}{t-2} dt.$$

Let

$$u = t - 2.$$

Differentiate:

$$du = dt.$$

Therefore,

$$\int \frac{1}{t-2} dt = \int \frac{1}{u} du.$$

Using

$$\int \frac{1}{u} du = \ln |u|,$$

we obtain

$$\ln |u|.$$

Substituting back $u = t - 2$,

$$\ln |t - 2|.$$

0.4.10 Step 10: Combine the Results

Adding the two antiderivatives,

$$y = -\ln |t - 1| + \ln |t - 2| + C.$$

0.4.11 Step 11: Combine the Logarithms

Use the logarithm rule

$$\ln(A) - \ln(B) = \ln\left(\frac{A}{B}\right).$$

Therefore,

$$\begin{aligned} & -\ln |t - 1| + \ln |t - 2| \\ &= \ln |t - 2| - \ln |t - 1| \\ &= \ln \left| \frac{t - 2}{t - 1} \right|. \end{aligned}$$

Thus,

$$y = \ln \left| \frac{t - 2}{t - 1} \right| + C.$$

0.4.12 Step 12: Apply the Initial Condition

The initial condition is

$$y(0) = 1.$$

Substitute $t = 0$ into the solution.

$$1 = \ln \left| \frac{0 - 2}{0 - 1} \right| + C.$$

Simplify the fraction:

$$\begin{aligned} 1 &= \ln \left| \frac{-2}{-1} \right| + C. \\ 1 &= \ln(2) + C. \end{aligned}$$

Solve for C .

$$C = 1 - \ln(2).$$

0.4.13 Step 13: Final Solution

Substitute the value of C into the general solution.

$$y(t) = \ln \left| \frac{t-2}{t-1} \right| + 1 - \ln(2).$$

Therefore,

$$\boxed{y(t) = \ln \left| \frac{t-2}{t-1} \right| + 1 - \ln(2)}.$$

0.5 Solution of Problem 4

Given:

$$t^2 y'' + 5ty' + 4y = 0, \quad y(1) = 0$$

Proposed solution:

$$y_1(t) = t^{-2} \ln t$$

0.5.1 Objective

Verify that $y_1(t) = t^{-2} \ln t$ satisfies both:

1. The differential equation

$$t^2 y'' + 5ty' + 4y = 0$$

2. The initial condition

$$y(1) = 0$$

0.5.2 Solution

Step 1: Compute the First Derivative

Given

$$y_1 = t^{-2} \ln t$$

Using the product rule,

$$\frac{d}{dt}(uv) = u'v + uv'$$

Let

$$u = t^{-2}, \quad v = \ln t$$

Then

$$u' = -2t^{-3}, \quad v' = \frac{1}{t}$$

Therefore,

$$y_1' = (-2t^{-3})(\ln t) + t^{-2} \left(\frac{1}{t} \right)$$

$$y_1' = -2t^{-3} \ln t + t^{-3}$$

Factoring out t^{-3} gives

$$y_1' = t^{-3}(1 - 2 \ln t)$$

Step 2: Compute the Second Derivative

Differentiate

$$y_1' = t^{-3}(1 - 2 \ln t)$$

Again using the product rule, let

$$u = t^{-3}, \quad v = (1 - 2 \ln t)$$

Then

$$u' = -3t^{-4}$$

and

$$v' = -\frac{2}{t}$$

Therefore,

$$y_1'' = (-3t^{-4})(1 - 2 \ln t) + t^{-3} \left(-\frac{2}{t} \right)$$

$$y_1'' = -3t^{-4}(1 - 2 \ln t) - 2t^{-4}$$

Expanding,

$$y_1'' = -3t^{-4} + 6t^{-4} \ln t - 2t^{-4}$$

Combining like terms,

$$y_1'' = t^{-4}(6 \ln t - 5)$$

Step 3: Substitute into the Differential Equation

The differential equation is

$$t^2 y'' + 5t y' + 4y$$

Compute each term separately.

First term:

$$\begin{aligned} t^2 y_1'' &= t^2 [t^{-4}(6 \ln t - 5)] \\ &= t^{-2}(6 \ln t - 5) \end{aligned}$$

Second term:

$$\begin{aligned} 5t y_1' &= 5t [t^{-3}(1 - 2 \ln t)] \\ &= 5t^{-2}(1 - 2 \ln t) \end{aligned}$$

Third term:

$$4y_1 = 4t^{-2} \ln t$$

Adding the three terms,

$$\begin{aligned} & t^2 y_1'' + 5t y_1' + 4y_1 \\ &= t^{-2}(6 \ln t - 5) + 5t^{-2}(1 - 2 \ln t) + 4t^{-2} \ln t \end{aligned}$$

Factor out t^{-2} :

$$= t^{-2} \left[(6 \ln t - 5) + 5(1 - 2 \ln t) + 4 \ln t \right]$$

Expand:

$$= t^{-2} \left[6 \ln t - 5 + 5 - 10 \ln t + 4 \ln t \right]$$

Combine like terms:

$$\begin{aligned} &= t^{-2} \left[(6 - 10 + 4) \ln t + (-5 + 5) \right] \\ &= t^{-2}(0) \\ &= 0 \end{aligned}$$

Hence,

$$t^2 y_1'' + 5t y_1' + 4y_1 = 0$$

Therefore, y_1 satisfies the differential equation.

Step 4: Verify the Initial Condition

Evaluate $y_1(1)$:

$$y_1(1) = 1^{-2} \ln(1)$$

Since

$$\ln(1) = 0$$

it follows that

$$y_1(1) = 0$$

Thus the initial condition is satisfied.

Conclusion

Since

$$t^2 y_1'' + 5t y_1' + 4y_1 = 0$$

and

$$y_1(1) = 0,$$

the function

$$y_1(t) = t^{-2} \ln t$$

satisfies both the differential equation and the initial condition.

Therefore,

$$\boxed{y_1(t) = t^{-2} \ln t}$$

is a solution of the given initial value problem.

0.6 Solution of Problem 5

0.6.1 Problem

Solve the differential equation

$$y' + 2y = \frac{e^{-2t}}{1 + t^2}.$$

0.6.2 Solution

This is a first-order linear differential equation.

Recall the standard linear form:

$$y' + P(t)y = Q(t).$$

Comparing with the given equation,

$$y' + 2y = \frac{e^{-2t}}{1 + t^2},$$

we identify

$$P(t) = 2,$$

and

$$Q(t) = \frac{e^{-2t}}{1 + t^2}.$$

Step 1: Find the Integrating Factor

The integrating factor is

$$\mu(t) = e^{\int P(t) dt}.$$

Substituting $P(t) = 2$,

$$\mu(t) = e^{\int 2 dt}.$$

Evaluating the integral,

$$\mu(t) = e^{2t}.$$

Therefore, the integrating factor is

$$\boxed{\mu(t) = e^{2t}}.$$

Step 2: Multiply the Differential Equation by the Integrating Factor

Multiply every term of the differential equation by e^{2t} :

$$e^{2t}y' + 2e^{2t}y = e^{2t} \left(\frac{e^{-2t}}{1+t^2} \right).$$

Simplifying the right-hand side,

$$e^{2t}e^{-2t} = 1.$$

Therefore,

$$e^{2t}y' + 2e^{2t}y = \frac{1}{1+t^2}.$$

Step 3: Recognize the Product Derivative

Recall the product rule:

$$\frac{d}{dt}(uv) = u \frac{dv}{dt} + v \frac{du}{dt}.$$

Let

$$u = e^{2t}, \quad v = y.$$

Then

$$\frac{du}{dt} = 2e^{2t}.$$

Applying the product rule,

$$\frac{d}{dt}(e^{2t}y) = e^{2t}y' + 2e^{2t}y.$$

This matches the left-hand side of the differential equation exactly.

Hence,

$$\frac{d}{dt}(e^{2t}y) = \frac{1}{1+t^2}.$$

Step 4: Integrate Both Sides

Integrating both sides with respect to t ,

$$\int \frac{d}{dt}(e^{2t}y) dt = \int \frac{1}{1+t^2} dt.$$

The left-hand side simplifies to

$$e^{2t}y.$$

Using the standard antiderivative

$$\int \frac{1}{1+t^2} dt = \tan^{-1}(t) + C,$$

we obtain

$$e^{2t}y = \tan^{-1}(t) + C.$$

Step 5: Solve for y

Multiply both sides by e^{-2t} :

$$y = e^{-2t}(\tan^{-1}(t) + C).$$

Therefore,

$$y(t) = e^{-2t}(\tan^{-1}(t) + C).$$

0.6.3 Answer

$$y(t) = e^{-2t}(\tan^{-1}(t) + C).$$

0.7 Solution of Problem 6**0.7.1 Problem**

Solve the differential equation

$$y' - \frac{1}{t} \sin(t^2) = -\frac{2}{t}y.$$

0.7.2 Step 1: Put the Equation into Standard Linear Form

Add $\frac{2}{t}y$ to both sides:

$$y' + \frac{2}{t}y = \frac{\sin(t^2)}{t}.$$

Compare with the standard linear form

$$y' + P(t)y = Q(t).$$

Thus,

$$P(t) = \frac{2}{t},$$

and

$$Q(t) = \frac{\sin(t^2)}{t}.$$

0.7.3 Step 2: Find the Integrating Factor

The integrating factor is

$$\mu(t) = e^{\int P(t) dt}.$$

Substituting $P(t) = \frac{2}{t}$ gives

$$\mu(t) = e^{\int \frac{2}{t} dt}.$$

Using

$$\int \frac{1}{t} dt = \ln |t|,$$

we obtain

$$\mu(t) = e^{2 \ln |t|}.$$

Using the identity

$$e^{\ln a} = a,$$

we get

$$\mu(t) = |t|^2.$$

For $t > 0$,

$$\boxed{\mu(t) = t^2}.$$

0.7.4 Step 3: Multiply the Equation by the Integrating Factor

Multiply every term by t^2 :

$$t^2 y' + 2ty = t \sin(t^2).$$

Notice that

$$\frac{d}{dt}(t^2 y) = t^2 y' + 2ty.$$

Therefore,

$$\frac{d}{dt}(t^2 y) = t \sin(t^2).$$

0.7.5 Step 4: Integrate Both Sides

Integrating both sides gives

$$t^2 y = \int t \sin(t^2) dt + C.$$

Step 4a: Use a Substitution

Let

$$u = t^2.$$

Differentiate:

$$du = 2t \, dt.$$

Solve for $t \, dt$:

$$t \, dt = \frac{du}{2}.$$

Substitute into the integral:

$$\int t \sin(t^2) \, dt = \frac{1}{2} \int \sin(u) \, du.$$

Step 4b: Evaluate the Integral

Recall the basic antiderivative

$$\int \sin(u) \, du = -\cos(u) + C.$$

Therefore,

$$\frac{1}{2} \int \sin(u) \, du = -\frac{1}{2} \cos(u).$$

Substituting back $u = t^2$ gives

$$\int t \sin(t^2) \, dt = -\frac{1}{2} \cos(t^2).$$

Hence

$$t^2 y = -\frac{1}{2} \cos(t^2) + C.$$

0.7.6 Step 5: Solve for y

Divide both sides by t^2 :

$$\begin{aligned} y(t) &= \\ \frac{C - \frac{1}{2} \cos(t^2)}{t^2}. \end{aligned}$$

Equivalent form:

$$\boxed{y(t) = \frac{C}{t^2} - \frac{\cos(t^2)}{2t^2}}.$$

0.8 Solution of Problem 7

0.8.1 Problem

Solve the differential equation

$$y' + \cot(t)y = e^t \cot(t).$$

0.8.2 Step 1: Verify That the Equation Is Linear

A first-order linear differential equation has the form

$$y' + P(t)y = Q(t).$$

Comparing with

$$y' + \cot(t)y = e^t \cot(t),$$

we identify

$$P(t) = \cot(t)$$

and

$$Q(t) = e^t \cot(t).$$

Since the equation is in standard linear form, we can use the integrating factor method.

0.8.3 Step 2: Find the Integrating Factor

The integrating factor is defined by

$$\mu(t) = e^{\int P(t) dt}.$$

Substituting $P(t) = \cot(t)$ gives

$$\mu(t) = e^{\int \cot(t) dt}.$$

Recall that

$$\cot(t) = \frac{\cos(t)}{\sin(t)}.$$

Therefore,

$$\int \cot(t) dt = \int \frac{\cos(t)}{\sin(t)} dt.$$

Let

$$u = \sin(t).$$

Then

$$du = \cos(t) dt.$$

Substituting,

$$\int \frac{\cos(t)}{\sin(t)} dt = \int \frac{1}{u} du.$$

Using the logarithm rule,

$$\int \frac{1}{u} du = \ln |u| + C.$$

Replacing u with $\sin(t)$,

$$\int \cot(t) dt = \ln |\sin(t)|.$$

Thus,

$$\mu(t) = e^{\ln |\sin(t)|}.$$

Using the property

$$e^{\ln(x)} = x,$$

we obtain

$$\mu(t) = |\sin(t)|.$$

On an interval where $\sin(t) > 0$, we take

$$\boxed{\mu(t) = \sin(t)}.$$

0.8.4 Step 3: Multiply the Entire Equation by the Integrating Factor

Multiply each term by $\sin(t)$:

$$\sin(t)y' + \sin(t)\cot(t)y = e^t \sin(t)\cot(t).$$

Using

$$\cot(t) = \frac{\cos(t)}{\sin(t)},$$

we find

$$\sin(t)\cot(t) = \sin(t)\frac{\cos(t)}{\sin(t)} = \cos(t).$$

Therefore,

$$\sin(t)y' + \cos(t)y = e^t \cos(t).$$

0.8.5 Step 4: Recognize the Product Rule

Recall the product rule:

$$\frac{d}{dt}[u(t)v(t)] = u'(t)v(t) + u(t)v'(t).$$

Let

$$u(t) = y$$

and

$$v(t) = \sin(t).$$

Then

$$\frac{d}{dt}[y \sin(t)] = y' \sin(t) + y \cos(t).$$

Notice that the left-hand side of our differential equation is exactly

$$\sin(t)y' + \cos(t)y.$$

Therefore,

$$\frac{d}{dt}[y \sin(t)] = e^t \cos(t).$$

0.8.6 Step 5: Integrate Both Sides

Integrate both sides:

$$\int \frac{d}{dt}[y \sin(t)] dt = \int e^t \cos(t) dt.$$

The left-hand side simplifies to

$$y \sin(t).$$

Thus,

$$y \sin(t) = \int e^t \cos(t) dt + C.$$

We now evaluate the integral

$$I = \int e^t \cos(t) dt.$$

0.8.7 Step 6: Evaluate the Integral $\int e^t \cos(t) dt$

Step 6a:

Let

$$I = \int e^t \cos(t) dt.$$

Using integration by parts,

$$\int u dv = uv - \int v du.$$

Choose

$$u = \cos(t), \quad dv = e^t dt.$$

Then

$$du = -\sin(t) dt, \quad v = e^t.$$

Substituting,

$$I = e^t \cos(t) - \int e^t (-\sin(t)) dt.$$

Therefore,

$$I = e^t \cos(t) + \int e^t \sin(t) dt.$$

Let

$$J = \int e^t \sin(t) dt.$$

Then

$$I = e^t \cos(t) + J.$$

Step 6b: Evaluate J

Again use integration by parts.

Choose

$$u = \sin(t), \quad dv = e^t dt.$$

Then

$$du = \cos(t) dt, \quad v = e^t.$$

Substituting,

$$J = e^t \sin(t) - \int e^t \cos(t) dt.$$

But

$$\int e^t \cos(t) dt = I.$$

Hence

$$J = e^t \sin(t) - I.$$

Step 6c: Substitute Back Into I

Recall

$$I = e^t \cos(t) + J.$$

Substitute $J = e^t \sin(t) - I$:

$$I = e^t \cos(t) + e^t \sin(t) - I.$$

Combine terms:

$$I = e^t (\cos(t) + \sin(t)) - I.$$

Add I to both sides:

$$2I = e^t (\cos(t) + \sin(t)).$$

Divide by 2:

$$I = \frac{e^t}{2} (\cos(t) + \sin(t)) + C.$$

Therefore,

$$\int e^t \cos(t) dt = \frac{e^t}{2} (\cos(t) + \sin(t)) + C.$$

0.8.8 Step 7: Substitute Back Into the Solution

We had

$$y \sin(t) = \int e^t \cos(t) dt + C.$$

Substituting the result,

$$y \sin(t) = \frac{e^t}{2} (\cos(t) + \sin(t)) + C.$$

0.8.9 Step 8: Solve for y

Divide both sides by $\sin(t)$:

$$y = \frac{\frac{e^t}{2} (\cos(t) + \sin(t))}{\sin(t)} + \frac{C}{\sin(t)}.$$

Separate the fraction:

$$y = \frac{e^t}{2} \left(\frac{\cos(t)}{\sin(t)} + \frac{\sin(t)}{\sin(t)} \right) + C \csc(t).$$

Since

$$\frac{\cos(t)}{\sin(t)} = \cot(t)$$

and

$$\frac{\sin(t)}{\sin(t)} = 1,$$

we obtain

$$y = \frac{e^t}{2} (\cot(t) + 1) + C \csc(t).$$

0.8.10 Final Answer

$$y(t) = \frac{e^t}{2} (1 + \cot(t)) + C \csc(t)$$